



西安交通大学
XI'AN JIAOTONG UNIVERSITY

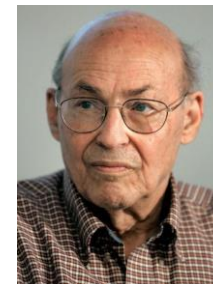
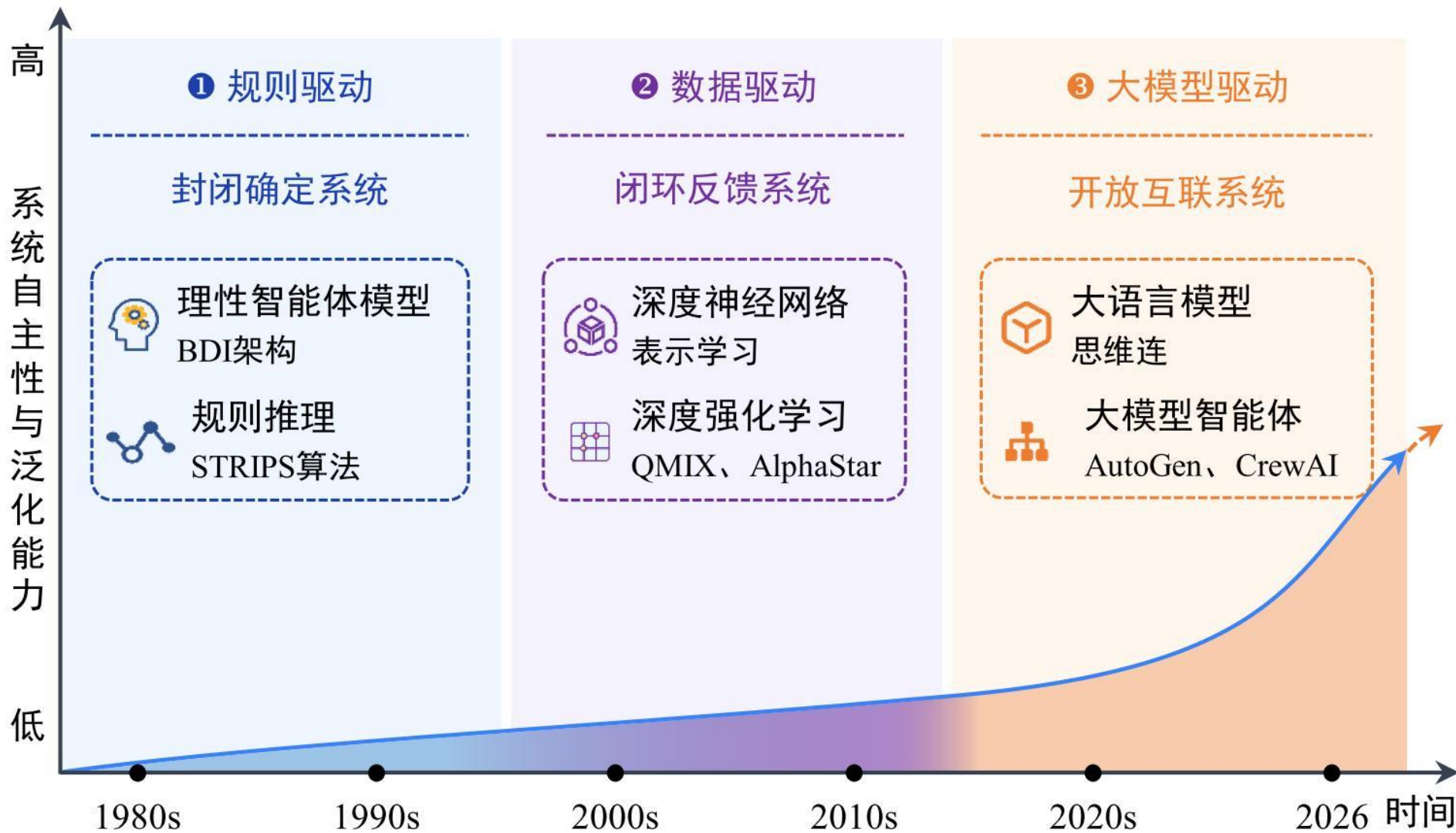


从单智能体到多智能体协同： 协作、归因与演化

马杰

智能网络与网络安全教育部重点实验室

智能体系统的三阶段发展趋势



Marvin Minsky

Society of Mind

做简单的事情，无需心智或思考



Michael Wooldridge

Intelligent agents

在环境中自主感知并执行动作

Agent 定义的变化



多智能体系统的群智是否可以涌现?

- **群体智能**：是一门研究自然系统与人工系统的学科，这些系统由众多个体构成，并通过**去中心化控制与自组织机制**实现协同运作。
- 研究表明^[1-3]：**多智能体性能与智能体的规模与多样性正相关！**扩大规模可拓宽解空间，增强多样性有助于提升搜索效率。

训练阶段尺度律（硅基）

性能与模型参数、训练数据、计算量近似幂率关系。



6个月100亿度

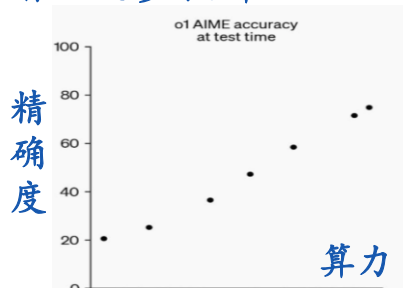
VS



1年100亿度

推理阶段尺度律（硅基）

性能随推理计算量、内存、推理时间的增加逐步提升。



多智能体尺度律（硅基）

80个蚂蚁协同完成“搬钢琴”谜题；如果是7个蚂蚁则不行^[3]。（碳基）



[1] Dorigo M, Birattari M, Stutzle T. Ant colony optimization. IEEE computational intelligence magazine, 2007.

[2] Tilman D, Wedin D, Knops J. Productivity and sustainability influenced by biodiversity in grassland ecosystems. Nature, 1996.

[3] Dreyer T, Haluts A, Korman A, et al. Comparing cooperative geometric puzzle solving in ants versus humans, PNAS 2025.

多智能体系统“生命周期”框架 (LIFE)

LIFE框架

演化

4 5

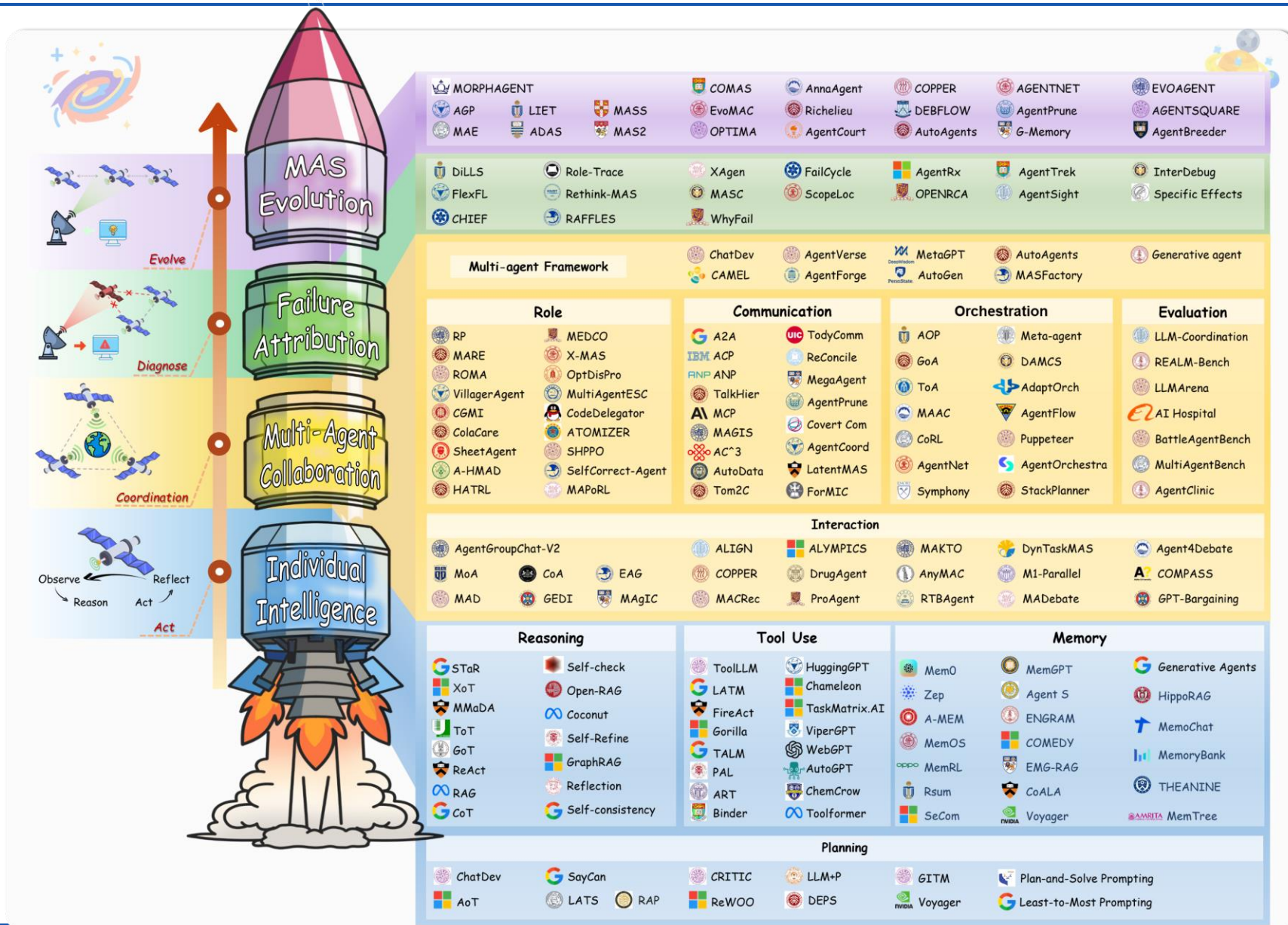
异常归因

多智能体
协作

2 3

单体智能

1



群体智能
基本原则

5 适应性

4 稳定性

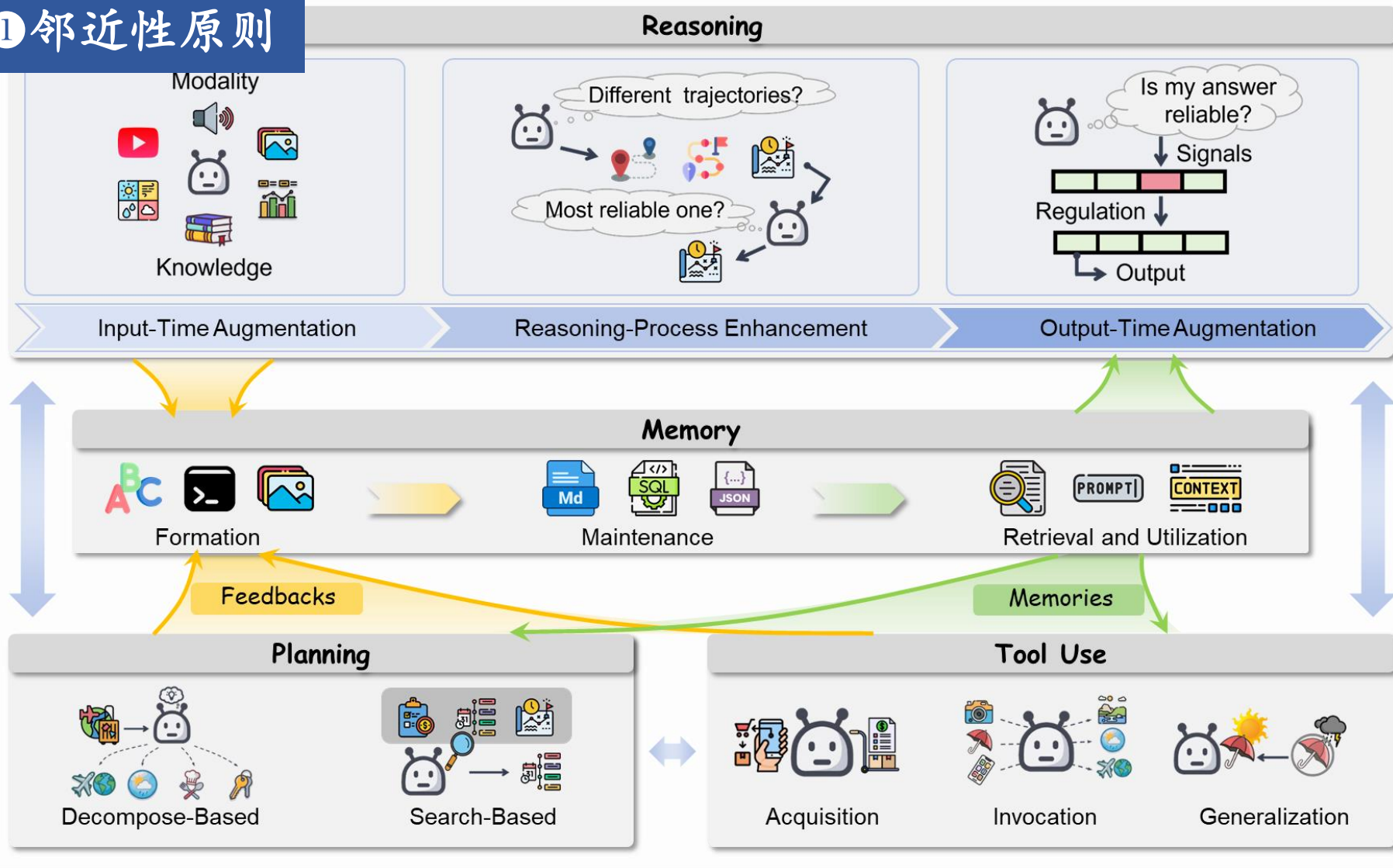
3 多样化
响应

2 质量

1 邻近性

单智能体能力框架 (Lay the capability foundation)

① 邻近性原则



推理决定智能体能否处理复杂指令；

记忆决定它能否利用历史经验；

规划决定它能否拆解长程任务；

工具使用能力则决定它能否突破模型自身的知识和执行边界。

多智能体协作框架 (Integrate agents through collaboration)

② 质量原则

Communication Protocol & Interface Design



- ✓ Establish data format and sharing rules

Role Engineering & Capability Design



- ✓ Define diverse agent competencies
- ✓ Engineer skillsets

Concept & Domain Modeling



- ✓ Understand goals
- ✓ Map environment requirements

Foundation

Role

Role Capabilities

Homogeneous



Generic Tourism Agents

Heterogeneous



Hotel Recommendation Agent



Weather Forecast Agent



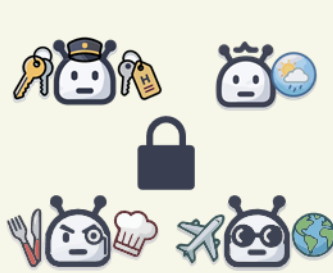
Restaurant Recommendation Agent



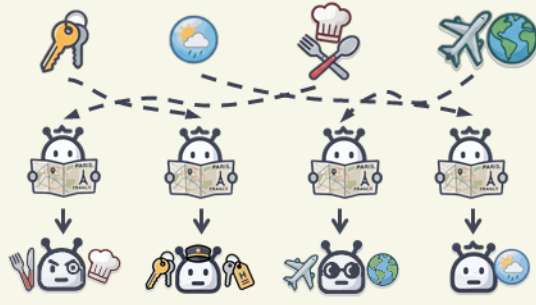
Attractions & Transportation Agent

Role Allocation

Static



Dynamic



Communication

Communication modes

Explicit

Requesting weather forecast for Paris next week.



Protocols

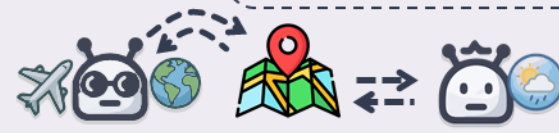
Forecast for Paris: Sunny, max 22°C



Implicit

Action: Update Shared Itinerary Map with Paris sightseeing plans.

1. Perceive: Noted Paris sightseeing entry on Oct 12th;
2. Infer: Need to provide Paris weather data;
3. Internal State Update: Update forecast for Paris on Oct 12th.



多智能体协作框架 (Integrate agents through collaboration)

③ 多样化响应原则

Performance Evaluation & Feedback



- ✓ Measure goal success
- ✓ Gather post-operation feedback

Deployment & System Runtime



- ✓ Activate agents
- ✓ Initiate real-time operations

Coordination Mechanisms & Topology Design



- ✓ Determine organizational structure and control logic

Execution

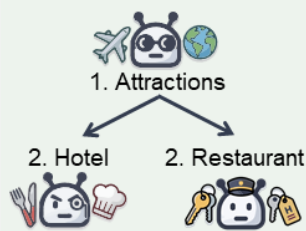
Interaction

Information Flow

Sequential

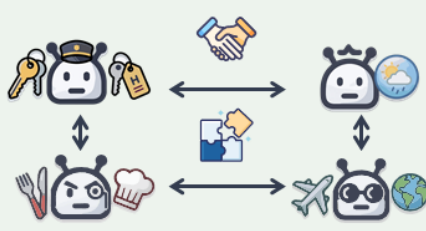


Parallel



Interaction Patterns

Cooperative



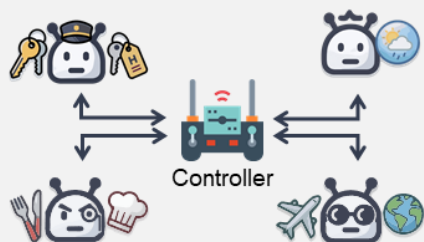
Competitive



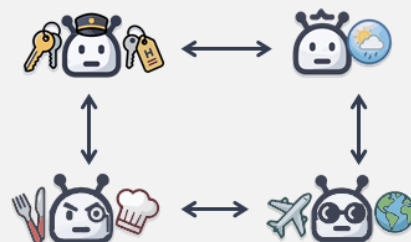
Structure

Orchestration Topology

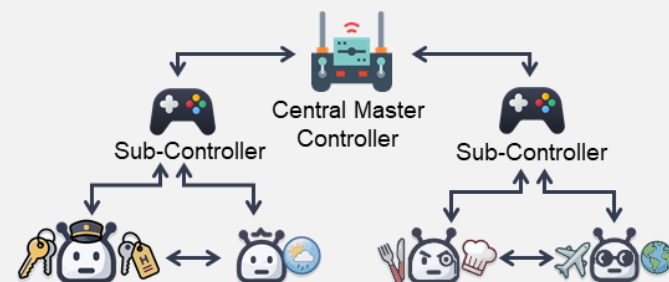
Centralized



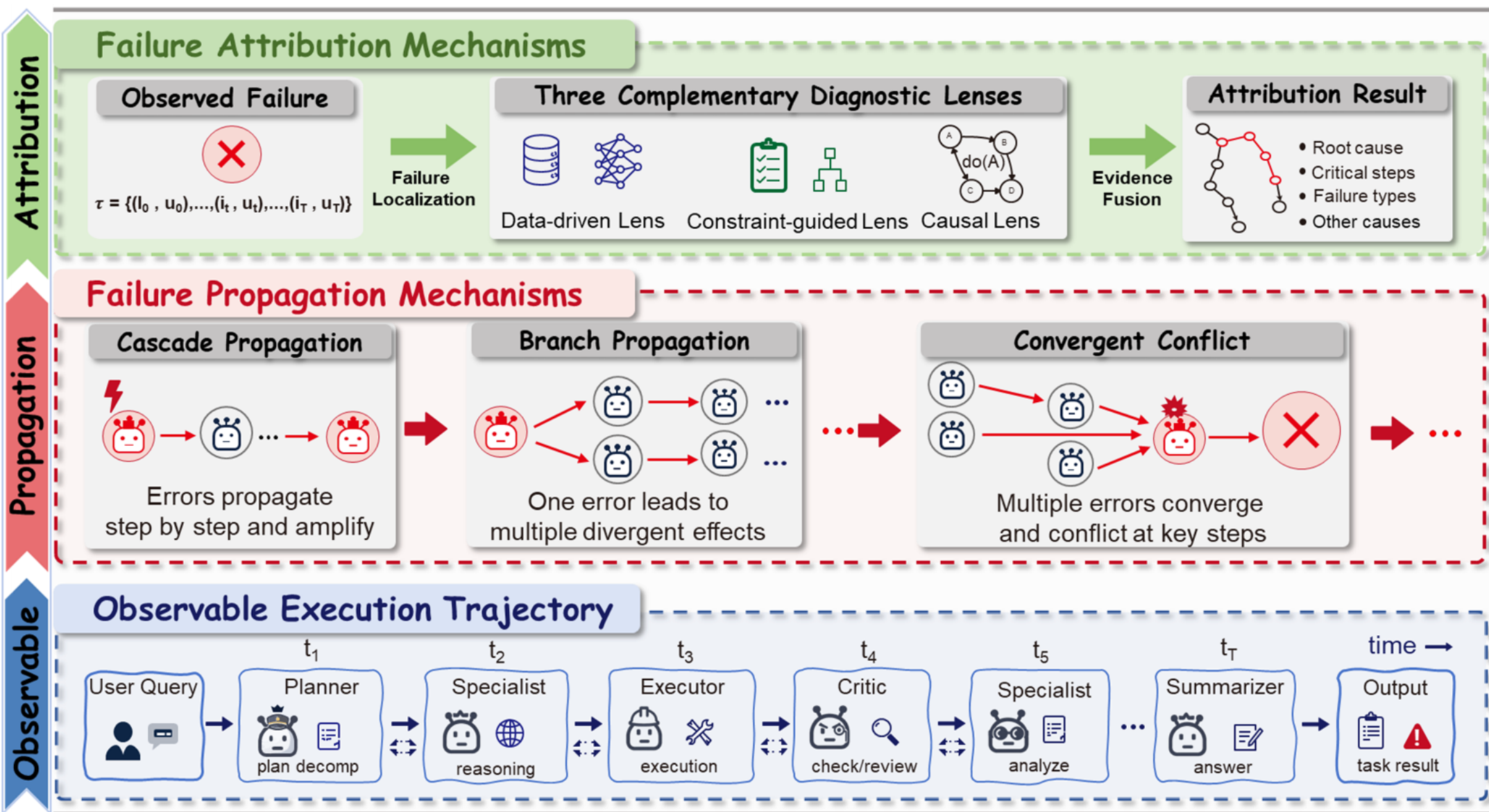
Distributed



Hybrid



多智能体归因框架 (Find faults through attribution)



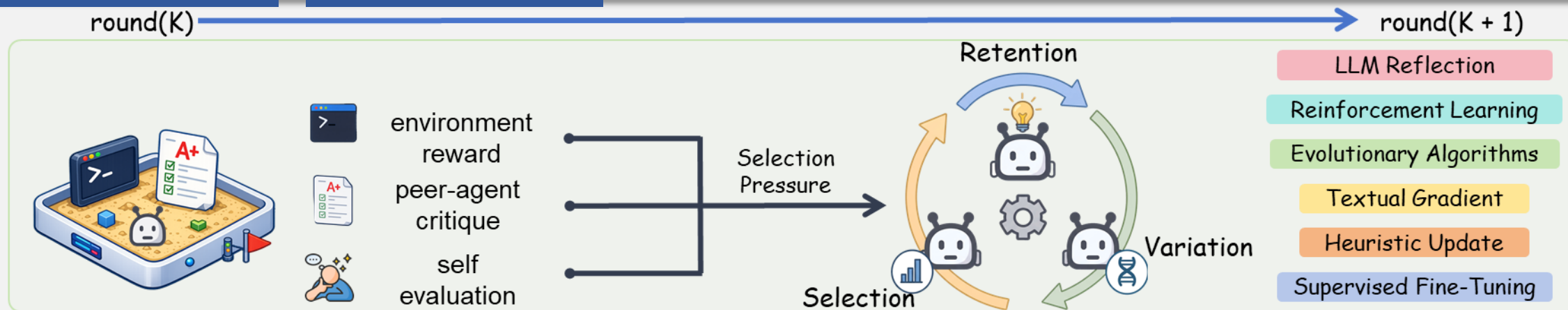
归因指系统
在多智能体
执行过程中
，对错误来
源、传播路
径与影响范
围进行定位
与分析的过
程。

多智能体自演化框架 (Evolve through self-improvement)

④ 适应性原则

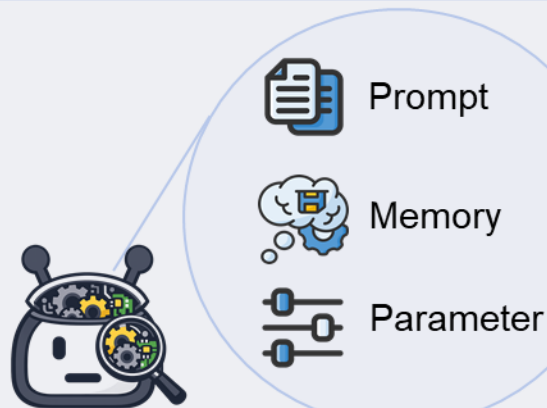
⑤ 稳定性原则

Self-Evolution

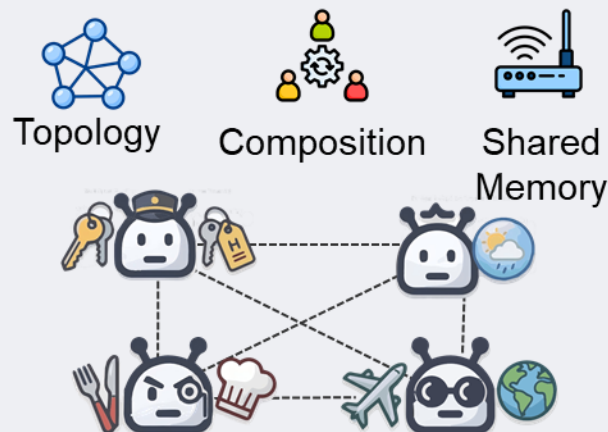


Targets

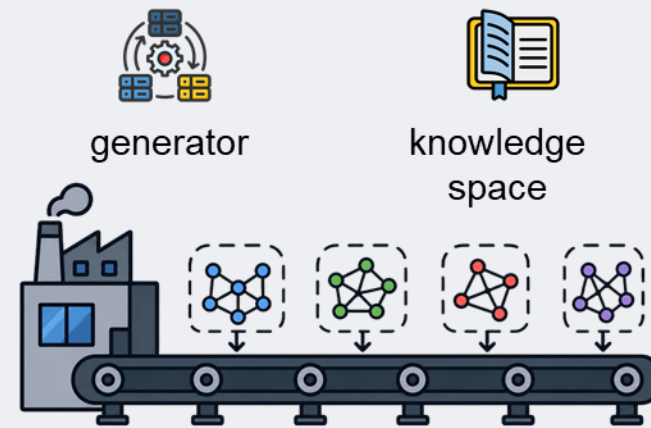
Agentic



Systemic



Meta





评测体系**不健全**

现有评测以结果为导向，缺乏效率（通信、计算）、长期（长程）稳定性等方面的考量。



协作结构**不灵活**

已有系统仍依赖人工设定角色、流程和通信方式。如何兼顾灵活性、协作效率、可靠性？



归因与演化**不可控**

归因目前侧重于事后分析，缺乏事中测评，以指导系统演化方向。如何确定局部和全局优化的契机，并朝着兼顾成本、稳定性、可控性的方向发展？



衷心感谢，敬请指正！

马杰

智能网络与网络安全教育部重点实验室